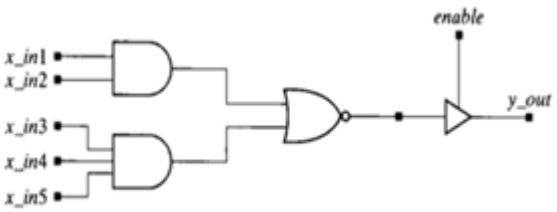
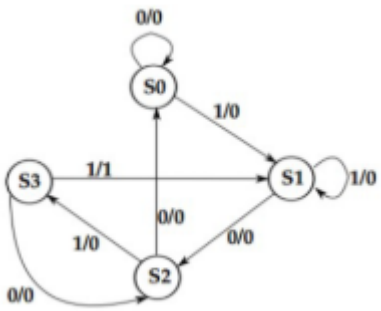
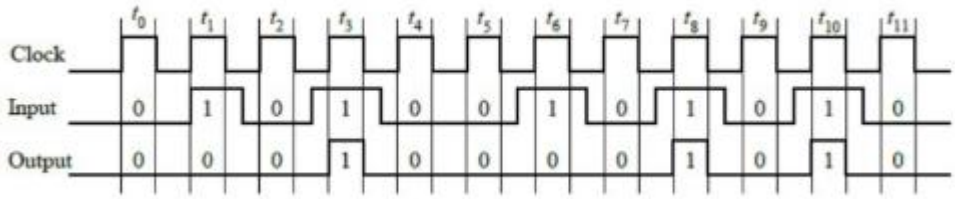
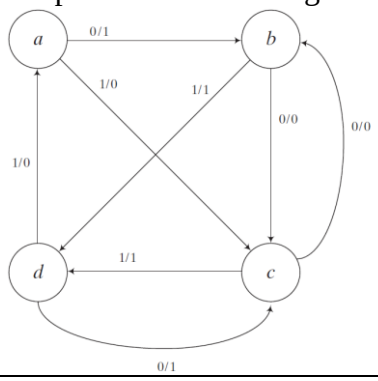
ADDC with HDL (EC234AI) **Tutorial Questions**	
Unit 1	
1.	Write truth table, for 4 bit gray code and write logic diagram and Verilog code to convert 4 bit binary number to gray code and gray code to binary using primitive XOR gates. a) Write block diagram of 4 bit BCD adder using 4bit parallel adders and gates. Write structural code for 4 bit parallel adder assuming full adders. Use full_adder component with a[3:0], b[3:0] and cin inputs, sum[3:0], cout as outputs.
2.	Write structural code for 4 bit BCD adder using 4bit parallel adders and gates. Use 4 bit parallel designed in (a) as component
3.	Write logic diagram for 3:8 decoder using 2:4 decoder and structural Verilog code for the 3 to 8 decoder using 2:4 decoder as component.
4.	a) Write data flow for the logic circuit shown in fig . Assume each gate has propagation delay of 5 units. b) Write structural flow for the logic circuit shown in fig. Assume each gate has propagation delay of 5 units. Also write test bench to chack any four passible input values. 
5.	Draw the logic diagram for the sequential circuit described by the following HDL module: **module** Seq_Ckt (**input** A, B, C, E **output reg** Q,input CLK,); **reg E;** **always @ (posedge** CLK) **begin** E <= A || B; Q <= E && C; **end** **endmodule**
Unit 2	
6.	A four-bit binary number is represented as A3A2A1A0, where A0 is the LSB. Design a logic circuit that will produce a HIGH output whenever the binary number is greater than 0010 & less than 1100.

7.	Design a combinational logic circuit that detects invalid BCD inputs. The circuit should produce an output of logic 1 when the 4-bit input does not represent a valid BCD digit and logic 0 for valid BCD inputs.
8.	Design a combinational circuit with three inputs, x, y, and z, and three outputs, A, B, and C, to perform the following operation: When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is two less than the input.
9.	Design a 2-digit BCD adder capable of adding two decimal numbers (00–99). Show carry correction logic.
10.	Design a logic circuit to realize the following function F (ABCD) = ∑(1,3,4,5,6,11,12,14) using a 4:1 MUX. Use CD as the select lines of the 4:1 MUX.
Unit 3	
11.	Design a counter that counts the sequence 0-2-3-1-0-2-3
12.	Write a verilog code for a synchronous counter that counts 0-1-3-2 with synchronous active high rst
13.	Construct a JK flip-flop using a D flip-flop, a two-to-one-line multiplexer, and an inverter.
14.	A PN flip-flop has four operations: clear to 0, no change, complement, and set to 1, when inputs P and N are 00, 01, 10, and 11, respectively. a) Tabulate the characteristic table. b) Derive the characteristic equation. c) Tabulate the excitation table. d) Show how the PN flip-flop can be converted to a D flip-flop.
15.	Using JK flip flops, design a synchronous counter for the following count sequence: 4-6-7-3-1----4-6……. (Avoid Lock out condition). Write behavioural modelling Verilog code for the same.
16.	Derive the state table, state diagram and write sequential circuit using JK FF for the expression given below A(t+1) = A’B’CD+A’B’C+ACD+AC’D’ B(t+1) = A'C+CD’+A’BC’ C(t+1) = B D(t+1) = D’
17.	Design a sequential circuit with two D flip-flops A and B, and one input x_in. 1. (a) When x_in=0, the state of the circuit remains the same. When x_in=1, the circuit goes through the state transitions from 00 to 01, to 11, to 10, back to 00, and repeats. 2. (b) When x_in=0, the state of the circuit remains the same. When x_in=1, the circuit goes through the state transitions from 00 to 11, to 01, to 10, back to 00, and repeats.
18.	A sequential circuit has two JK flip-flops A and B, two inputs x and y, and one output z. The flip-flop input equations and circuit output equation are JA=Bx+B′y′ ; KA=B′xy′ ; JB=A′x ; KB=A+xy′ ; z=Ax′y′+Bx′y′

	(a) Draw the logic diagram of the circuit. (b) Tabulate the state table. (c) Derive the state equations for A and B. (d) Draw the state diagram
19.	A sequential circuit has two *JK* flip-flops *A* and *B* and one input *x* . The circuit is described by the following flip-flop input equations: $JA = x$ $KA = B$ $JB = x$ $KB = A_$ (a) Derive the state equations $A(t + 1)$ and $B(t + 1)$ by substituting the input equations for the *J* and *K* variables. (b) Draw the state diagram of the circuit.
20.	Design of 3-bit Gray code counter using JK flip-flops, include excitation table, K-map simplification and final logic.. Write a behavioural Verilog code to implement the same.
21.	Design a 3-bit synchronous up/down counter using jk flipflops. The counter has mode control input M; when M=0, counter performs up counting and when M=1, counter performs down counting. write a verilog code for the same.
	Unit 4
22.	Construct the logic diagram of a 4-bit register with four *D flip-flops* and 4×1 multiplexers with mode selection input m1 and m0. The register operates according to the following function table: m1 | m0 | Register operation 0 | 0 | No Change 0 | 1 | Complement the four Output 1 | 0 | Clear register to 0 (Synch) 1 | 1 | Load parallel data If data bits 1011 is loaded in the above logic, draw the timing diagram for 6 clock pulses. Develop Verilog code for the above shift register.
23.	Draw the state diagram for detecting the sequence 1001 using a Moore FSM (Non Overlapping). Write the Verilog code to implement the same using behavioral modelling.
24.	Write Verilog code for an 8-bit serial adder using shift registers and a carry flip-flop. Draw a Block diagram for the same.
25.	Consider the case of a circuit to detect a pair of 1's or 0's in the single bit input. That is, input will be a series of one's and zero's. If two one's or two zero's comes one after another, output should go high. Otherwise output should be low. Develop a verilog Code for the circuit using Mealy FSM .
26.	Write a state table and Verilog code (Mealy FSM) for the state diagram shown in figure.

	
27.	For the given timing diagram identify the sequence which is detected and write a Verilog code for the detected sequence using mealy FSM, assuming that overlapping of pattern is allowed. 
28.	Distinguish between a synchronous preset and an asynchronous preset capability in a counter circuit. Discuss the relative advantages and disadvantages of each method. Design a 4-bit synchronous presettable counter that counts binary states 3,4,5,6,7,8 using IC74192. Draw the block diagram and explain the operation.
29.	Draw the state diagram of a circuit that detects the following sequence: 10110. The detector must assert output ‘z=1’ when the sequence is detected.(With Mealy Overlap and non overlap Model).
30.	Design a serial 2’s complementer with a shift register and a flip-flop. The binary number is shifted out from one side and it’s 2’s complement shifted into the other side of the shift register
31.	Design a 4 bit bidirectional shift register . Write logic diagram and explain the shift left and shift right function with an example. Use 2:1 MUX
32.	Write a parameterized Verilog module for an N-bit shift register with serial input and parallel output. Write suitable testbench to verify the functionality.
33.	Write a Verilog model of the Mealy FSM described by the state diagram in Fig. P Develop a test bench and demonstrate that the machine state transitions and output correspond to its state diagram. 

Unit 5

34. In a status register, explain the behaviour of the Zero flag when subtraction (A − B) is performed for the conditions (i) A > B and (ii) A ≥ B. Also, write the Boolean condition associated with the Zero flag.

35. For a binary subtraction operation F=A-B, Determine the status of all flags for i) A = B, ii) A ≥ B, iii) A < B iv) A ≤ B.

36. There are 4 registers with 4-bit size in a processor unit. With neat diagram show how data from register is transferred to ALU through a Bus (8 bit) and how result is stored in a register through Bus (8 bit)

37. A processor unit employs a scratchpad memory which has 64 rows with each row 8-bit size.

Write block diagram.

a) What is the size of the scratchpad memory?
b) How many lines are needed for the address?
c) How many lines are there for the input data?
d) What is the size of the MUX that selects between the input data and the output of the shifter?

38. Specify the control word that must be applied to the processor to implement the following micro-operations: Assume processor has 8 registers, 3 select lines for ALU, 1 Cin, and 3 select lines for shifter.

(a) $R_3 \leftarrow R_4 + R_5$
(b) $R_6 \leftarrow \bar{R}_6$
(c) $R_1 \leftarrow \text{shr } R_1$
(f) $R_2 \leftarrow \text{clc } R_2$
(g) $R_3 \leftarrow R_4 \oplus R_5$

39. With an example for each condition in the table shown, explain the effect of Cout and overflow flag.

Function select			Arithmetic function	$C_{out} = 1$ if
s_1	s_0	C_{in}		
0	0	0	$F = A$	
0	0	1	$F = A + 1$	$A = 2^n - 1$
0	1	0	$F = A + B$	$(A + B) > 2^n$
0	1	1	$F = A + B + 1$	$(A + B) > (2^n - 1)$
1	0	0	$F = A - B - 1$	$A > B$
1	0	1	$F = A - B$	$A \geq B$
1	1	0	$F = A - 1$	$A \neq 0$
1	1	1	$F = A$	

40. Design an arithmetic and logic unit using Full adder to perform the operations shown in table. Obtain the expression for inputs Xi, Yi and Zi of full adder. Write the logic diagram for one stage. Show how AND and OR operations are executed. Also show how addition and data transfer instructions are executed.

Arithmetic Operations					Logical Operations			
S_2	S_1	S_0	C_{in}=0	C_{in}=1	S_2	S_1	S_0	C_{in}=X
0	0	0	F=A	F=A+1	1	0	0	OR

0	0	1	F=A+B	F=A+B+1	1	0	1	XOR
0	1	0	F=A-B-1	F=A-B	1	1	0	AND
0	1	1	F=A-1	F=A	1	1	1	Complement